Fresh Solutions: a reproducible adapter harness for applied metaheuristic benchmarks

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November 3, 2026

Abstract

Applying metaheuristics to hard combinatorial problems is often slowed less by the choice of metaheuristic than by the engineering around it: problem-specific move representations, exact incremental state, replayable evidence, stopping rules, and benchmark bookkeeping are usually rebuilt for each project. Fresh Solutions provides an adapter contract and a reproducible benchmark harness, together with two reference implementations: quadratic-assignment robust tabu search and travelling-salesman iterated Lin–Kernighan. We evaluate the references on QAPLIB and TSPLIB and report the full distribution, including failures. Small and medium QAP instances are strong; the broad QAP panel has a heavy tail. Four TSP instances reach the known optimum or a 0.383% gap with iterated Lin–Kernighan; the internal method does not scale to the largest TSPLIB tail. An optional external LKH backend is reported separately. The contribution is the harness, exact replay contract, and reference adapters, not a general-framework experiment or a state-of-the-art solver claim.

1 Introduction

Operations-research practitioners rarely need a new metaheuristic. They need a reliable way to apply known methods to a new problem, preserve exact feasibility and objective accounting, and produce evidence that another person can replay. That work is repetitive and easy to get subtly wrong. Problem-specific move representations, incremental state, stopping rules, benchmark harnesses, seed handling, and evidence records are rebuilt for each project.

Fresh Solutions is a response to that engineering problem. It provides a problem-adapter contract, exact resident state and reversible moves, local and population search paths, checkpointing, exact replay, convergence records, and public-library reference implementations for QAP and TSP. The paper asks a narrow engineering question: can a user add a problem-specific neighbourhood or reference implementation and obtain credible, replayable evidence without rebuilding the surrounding machinery? It does not claim that the generic engine has been tested as the source of the headline results.

2 Related work

The quadratic assignment problem asks for a permutation that minimises a flow–distance product. Robust tabu search is a long-standing baseline for QAPLIB [Burkard et al., 1997]. Taillard's robust tabu search remains a standard reference implementation [Taillard, 1991]; our QAP adapter follows the same move-and-memory idea, with exact swap deltas and an optional delta-matrix path.

The travelling-salesman problem has an equally long heuristic literature. The Lin–Kernighan variable-depth exchange [Lin and Kernighan, 1973] and Helsgaun's LKH implementation [Helsgaun,

2000] are the standard strong baselines. We implement a bounded candidate-list LK-style exchange and use external LKH only as an optional capability reference, not as part of the package. TSPLIB [Reinelt, 1991] provides the public instances.

Reproducibility in applied optimisation depends on more than a seed. It depends on the compiled model, candidate order, effect table, objective weights, hardware, and replay procedure. The harness stores those elements with each record and replays the stored schedule through the exact evaluator. The design is deliberately closer to a benchmark harness than to a new metaheuristic.

3 Framework

3.1 Adapter contract

An adapter supplies variables, options, exact and batched evaluation, feasibility, mutation and crossover, and reversible moves. The engine sees only this interface. A domain-specific method can be added as a separate module without changing the core.

3.2 Exact state and replay

Every reported incumbent is a complete assignment. Evaluation is exact, and the stored assignment must re-evaluate to the stored objective. The benchmark scripts do not trust the optimizer's internal counter; they replay the result through the adapter. A record with a mismatched schedule hash or replayed objective is not evidence.

3.3 Search layers

The harness exposes exact local search, population search with crossover, mutation, and elitism, and a hybrid handoff from a population elite to exact local refinement. The experiments in this paper use the local layer plus adapter-specific neighbourhoods. The population layer remains available but is not the source of the strongest results, and we do not claim a generic-engine framework experiment.

3.4 Evidence protocol

A reportable result names the pinned solver version or commit, the public instance and known or best-known value, the seed and budget, the stopping reason, the exact-replay result, and the wall time. Publication-facing campaigns use a patience rule; safety-cap hits are recorded as non-converged.

4 QAP evidence

The QAP reference implementation uses robust tabu search over facility swaps. The objective is

$$\min_{\pi \in \Pi_n} \sum_{i=1}^{n} \sum_{j=1}^{n} F_{ij} D_{\pi(i)\pi(j)}, \tag{1}$$

where F is the flow matrix and D the distance matrix. The adapter uses exact swap deltas; the accelerated path uses Taillard's delta-matrix update for symmetric zero-diagonal instances and a Numba general scan for asymmetric instances.

Table 1: QAP robust tabu, five seeds. The certified budget is the smallest budget satisfying the 0.10-point doubling rule.

Instance	Certified budget	Mean gap	Worst gap	Next doubling
`tai12a`	10,000	0.000%	0.000%	optimum on all seeds
`tai20a`	10,000	0.403%	0.470%	20k: 0.370%
`tai25a`	20,000	0.685%	0.850%	40k: 0.592%
`tai30a`	10,000	0.792%	1.139%	20k: 0.700%

Table 2: Large QAPLIB instances, five random starts, all stopping on patience. SciPy FAQ is an independent external baseline with the same five-seed protocol.

Instance	Fresh best	Fresh mean	SciPy mean	Fresh worst
`esc128`	0.000%	1.250%	12.500%	3.125%
`tai150b`	2.008%	2.976%	1.861%	3.699%
`tai256c`	0.350%	0.413%	120.481%	0.509%
`tho150`	0.166%	0.507%	1.523%	0.902%

4.1 Certified QAP panel

Table 1 reports the certified budget and five-seed gaps for the initial QAPLIB panel. The certified budget is the smallest budget whose next doubling improves the five-seed mean by no more than 0.10 percentage points. This is a stability rule, not a convergence proof.

4.2 Broad panel and large instances

Across 134 parseable QAPLIB instances and five seeds, 552 of 670 runs stopped on patience and 318 hit the known optimum. The converged subset has median gap 0.000% and mean 1.823%, with a worst case of 36.4%. The mean is pulled by a small number of hard outliers; the median is not evidence that the tail is solved.

Table 2 reports the large-instance multi-start results. The best-of-five values are not a portfolio-improvement claim; the mean values show that the multi-start result is better on `tai256c` and `tho150` but worse on `esc128` and `tai150b`. SciPy FAQ is a weak, fast external baseline; it wins `tai150b` and loses the other three on mean gap.

5 TSP evidence

The TSP reference implementation uses two layers: Or-opt/2-opt for construction and a bounded candidate-list Lin–Kernighan-style exchange with deterministic double-bridge kicks. For a tour π, the objective is

$$L(\pi) = \sum_{i=1}^{n} d_{\pi(i),\pi(i+1)}, \qquad \pi(n+1) = \pi(1). \tag{2}$$

5.1 Iterated Lin–Kernighan

Table 3 reports a four-instance experiment starting from converged Or-opt tours. The 100,000-kick patience budget is extra work relative to Or-opt; the comparison is quality-at-additional-cost, not an equal-budget ablation.

Table 3: TSP iterated Lin–Kernighan from converged Or-opt tours, five seeds.

Instance	Or-opt mean gap	ILK mean gap	ILK worst gap
`berlin52`	6.433%	0.000%	0.000%
`kroA100`	7.323%	0.000%	0.000%
`kroB100`	3.696%	0.000%	0.000%
`ch150`	8.229%	0.383%	0.475%

A kicks ablation separates the effect: LK-only means 0.780–2.580%, ILK-5 means 0.000–0.558%, and ILK-20 means 0.000–0.383%. The double-bridge kicks are a material part of the result, not a cosmetic restart.

5.2 Broad panel and the LKH boundary

Across 78 TSPLIB instances with $n \leq 1000$, 236 of 390 runs stopped on patience and 183 hit the known optimum. The converged subset has median gap 0.000% and mean 0.100%. The large tail is weak: only 3 of 100 runs above $n = 300$ reached patience, and 97 hit the safety cap. The internal method is therefore a small/medium TSP solver.

External LKH, at its stated budget and configuration, reaches 71 of 100 optima on the same large tail, mean gap 0.092%, median 0%, and worst 1.930%. LKH is optional external research-use software, not bundled and not claimed as Fresh Solutions' own heuristic. The comparison is capability-only, not matched-time.

6 Applied case

TSA24/Prince George is a public-data applied case. The frozen build, target, and seed are published as `ws4-tsa24-evenflow-v1` in `https://github.com/UBC-FRESH/femic-tsa24-instance`, derived from public BC VRI, FADM, and SiteProd data. The even-flow target is a WS4 scenario configuration, not an official TSA decision.

The 1,000-generation GPU-GA record has objective 8,588,725.370498; the 10,000-generation same-engine baseline has 76,540.740733; the 100,000-generation five-seed campaign spans 512.574–518.621 (mean 515.682), with exact replay. The CPU-SA independent-search baseline is 335,044,976.675435 and is time-limited, not converged. No partner-impact claim is made.

7 Discussion and limitations

The harness makes several engineering properties explicit: the adapter boundary, exact resident state, replay, convergence records, and benchmark provenance. Those are useful even when the underlying heuristic is standard. The strongest QAP and TSP results come from adapter-specific modules (`qap_tabu.py` and `tsp_lk.py`), not from the generic engine. A generic-framework experiment remains open.

The QAP median of 0% is dominated by small instances; the mean and tail are the honest summary. The large-QAP SciPy baseline is weak and mixed. The TSP ILK result is four instances; the 78-instance panel uses nearest-neighbour starts and a different experiment. The internal TSP path does not scale to the largest TSPLIB tail; external LKH is the strong path there. The applied case is public-data, not partner-endorsed.

Patience is a plateau rule, not global optimality. LKH is external and research-use. The evidence release and tag remain pending. No state-of-the-art or universal-dominance claim is made.

8 Conclusion

Fresh Solutions provides a reproducible benchmark harness and adapter boundary, with two reference metaheuristic implementations and exact replay. It is useful today for small and medium QAP/TSP work and for building applied cases. It is not yet a general-framework experiment or a universal large-instance solver, and the paper says so.

Data and code availability

QAPLIB and TSPLIB instances are public. The harness, reference implementations, and evidence are in the Fresh Solutions papers repository. The TSA24 artifact is available at `https://github.com/UBC-FRESH/femic-tsa24-instance/releases/tag/ws4-tsa24-evenflow-v1`.

References

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